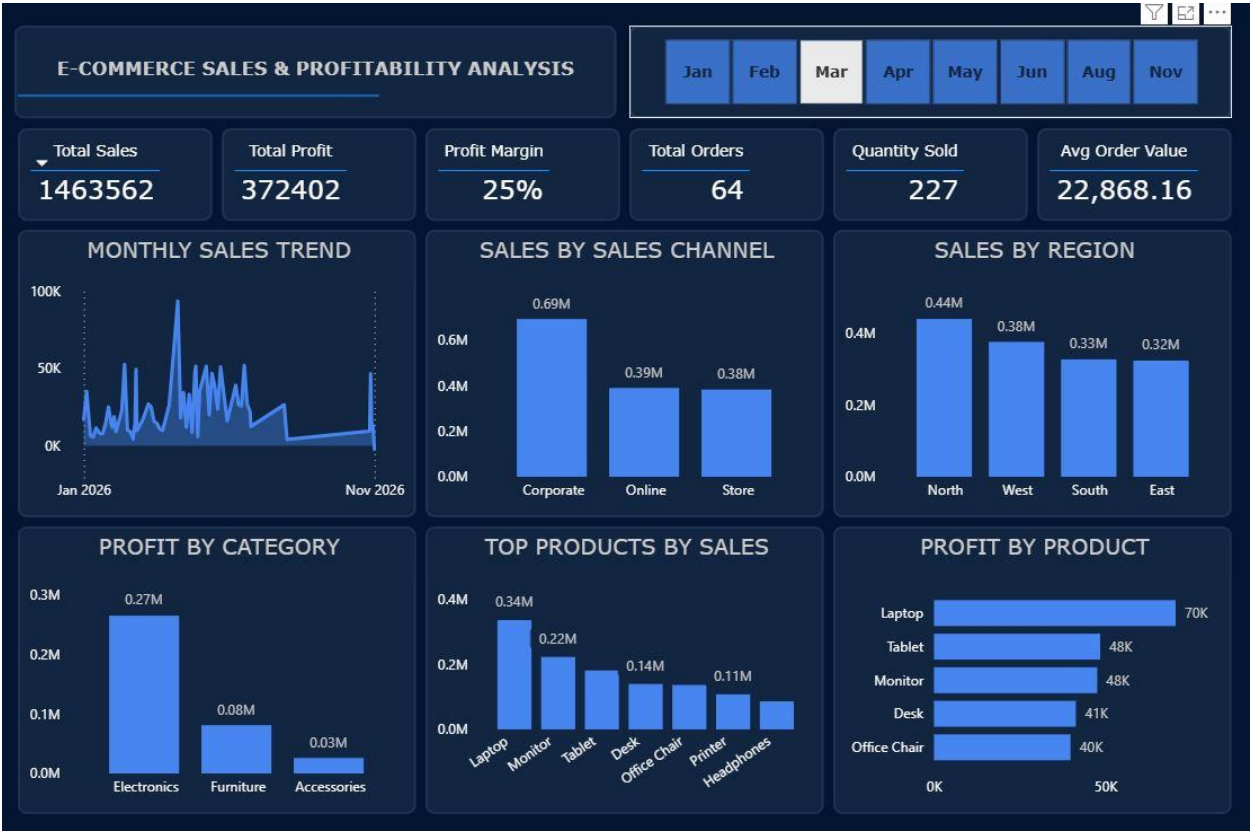


E-COMMERCE SALES & PROFITABILITY ANALYSIS

End-to-End Data Analytics Project

Data Analysis & Business Insights

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Tools Used: Excel | Power Query | SQL | Python/Pandas | Power BI | DAX

1. PROJECT OVERVIEW

Project Objective

The objective of this project was to analyze e-commerce sales and profitability data and convert raw, inconsistent business data into meaningful insights for decision-making.

The project focused on:

- Sales performance
- Profitability
- Product performance
- Discount impact
- Customer and transaction data
- Sales forecasting
- Business recommendations

The project followed an end-to-end data analytics workflow, using different tools for different purposes rather than performing the same analysis repeatedly in every tool.

Analytical Workflow

Excel → Power Query → Power BI Data Model → SQL → Python/Pandas → Power BI & DAX → Business Insights & Recommendations

2. DATASET STRUCTURE

The project used four main datasets:

1. Raw Sales Data

Contained transaction-level information such as:

- Order_ID
- Customer_ID
- Order_Date

- City
- Region
- Product_ID
- Product
- Category
- Quantity
- Unit_Price
- Discount
- Sales_Amount
- Cost_Amount
- Payment_Method
- Sales_Channel
- Salesperson

2. Customer Master

Contained:

- Customer_ID
- Customer_Name
- City
- Region
- Customer_Type
- Join_Date
- Age

3. Product Master

Contained:

- Product_ID
- Product
- Category
- Base_Price
- Supplier

4. Monthly Targets

Contained:

- Month
- Month Number
- Region
- Target

The target data was transformed from monthly columns into a normalized structure using Power Query.

3. DATA PROBLEMS IDENTIFIED

The raw data contained several realistic data-quality problems.

Date Problems

- Multiple date formats were present.
- Some dates were not initially recognized consistently.
- An invalid date such as **31/06/2026** was present.
- Dates needed to be converted into proper Date data types.

Numeric Problems

- Currency values contained symbols and commas, such as ₹52,000.
- Some numeric fields contained text.

- Discount values needed to be converted into a consistent numeric format.
- Invalid and negative values required review.

Quantity Problems

The Quantity field contained invalid entries such as:

- two
- Negative quantity values

These values required data-quality review.

Text Inconsistencies

The same business values appeared in different forms.

Examples:

- Kolkata / Kolkata
- East / east / EAST
- Electronics / electronics / ELECTRONICS
- Credit Card / credit card
- Online / online

Without standardization, these values could be treated as different categories during analysis.

Missing and Duplicate Data

The dataset was checked for:

- Missing Order IDs
- Duplicate Order IDs
- Missing important fields
- Invalid records

A record with a missing Order ID was removed after review.

4. DATA CLEANING — POWER QUERY

Power Query was used as the main ETL and data-cleaning tool.

Date Transformation

Mixed date formats were handled using appropriate locale-based conversion.

The final Order_Date field was converted into a proper Date data type.

Unit Price Cleaning

Currency symbols and commas were removed and the values were converted into numeric values.

For example:

₹52,000 → 52000

Discount Cleaning

Discount values were converted into numeric values.

Invalid values such as N/A were handled as null where appropriate.

Cost Amount Cleaning

Cost values were cleaned and converted into numeric values.

Missing Order ID

The record containing a missing Order_ID was removed.

Text Standardization

Text fields such as:

- City
- Region
- Category
- Payment Method
- Sales Channel

were standardized so that different capitalization and unnecessary spaces did not create duplicate categories.

5. DATA INTEGRATION

The cleaned sales data was enriched using the master datasets.

Customer Merge

Sales_Data was merged with Customers_Data using:

Customer_ID

The following customer attributes were added:

- Customer_Name
- Customer_Type
- Join_Date
- Age

Product Merge

Sales_Data was merged with Products_Master using:

Product_ID

The following product attributes were added:

- Product
- Category
- Base_Price
- Supplier

This created a richer analytical dataset while keeping sales transactions as the central business data.

6. TARGET DATA TRANSFORMATION

The Monthly Targets dataset originally stored each month as a separate column.

Power Query was used to **unpivot** these monthly columns.

The final structure became:

Region Month Target

East	Jan	Target
------	-----	--------

East	Feb	Target
------	-----	--------

...	...	...
-----	-----	-----

A Month Number field was also maintained to support chronological month ordering.

7. FINAL DATASET

After the cleaning and transformation process, the final sales dataset used for the analytical workflow contained:

64 transaction records

and

22 columns

The final analytical columns included:

Order_ID, Customer_ID, Order_Date, City, Region, Product_ID, Product, Category, Quantity, Discount, Sales_Amount, Unit_Price, Cost_Amount, Payment_Method, Sales_Channel, Salesperson, Customer_Name, Join_Date, Customer_Type, Age, Base_Price, Supplier.

The final dataset was then used across the subsequent analytical stages.

8. POWER BI DATA MODEL

The cleaned data was loaded into Power BI.

A dedicated Date table was created to support time-based analysis and forecasting.

Date Table

The Date table covered the complete range between the earliest and latest sales dates.

The relationship was created as:

Date_Table[Date] → Sales_Data[Order_Date]

Relationship:

One-to-Many

- One: Date_Table
- Many: Sales_Data
- Cross-filter direction: Single
- Relationship: Active

This structure allowed Power BI to work with a continuous calendar rather than only the dates on which transactions occurred.

9. POWER BI DAX MEASURES

The following measures were created for the dashboard.

Total Sales

Total Sales =

```
CALCULATE(  
    SUM(Sales_Data[Sales_Amount]),  
    ALL(Sales_Data[Order_Date])  
)
```

Total Profit

Total Profit =

```
SUM(Sales_Data[Sales_Amount]) -
```

SUM(Sales_Data[Cost_Amount])

Profit Margin

Profit Margin =

DIVIDE(
[Total Profit],
[Total Sales]
)

Total Orders

Total Orders =

DISTINCTCOUNT(Sales_Data[Order_ID])

Total Quantity Sold

Total Quantity Sold =

SUM(Sales_Data[Quantity])

Average Order Value

Average Order Value =

DIVIDE(
[Total Sales],
[Total Orders]
)

Profit

Profit =

SUM(Sales_Data[Sales_Amount]) -
SUM(Sales_Data[Cost_Amount])

10. SALES & PROFITABILITY DASHBOARD



The dashboard was designed to provide a compact overview of business performance.

KPI Section

The dashboard contains six key indicators:

1. **TOTAL SALES**
2. **TOTAL PROFIT**
3. **PROFIT MARGIN**
4. **TOTAL ORDERS**
5. **QUANTITY SOLD**
6. **AVG ORDER VALUE**

The KPI section provides management with a quick view of overall business performance.

Dashboard Visuals

Daily Sales Trend

Shows sales movement over individual transaction dates.

Purpose:

- Identify sales fluctuations
- Understand sales patterns over time
- Support time-based analysis

Sales by Region

Shows the contribution of different regions to total sales.

Purpose:

- Compare regional performance
- Identify stronger and weaker markets

Profit by Category

Shows profit contribution by product category.

Purpose:

- Identify profitable categories
- Compare category-level profitability

Top Products by Sales

Shows products ranked according to sales.

Purpose:

- Identify major revenue-driving products

Profit by Product

Shows product-level profit contribution.

Purpose:

- Compare revenue generation with actual profit contribution

Sales by Sales Channel

Shows sales across different sales channels.

Purpose:

- Understand channel contribution
- Support channel-level business decisions

Dashboard Filter

A **Month slicer** was added using:

Sales_Data[Month]

This allows the sales visuals to be filtered by month.

11. OVERALL FINANCIAL RESULT

SQL analysis of the cleaned dataset produced:

Metric	Result
Total Sales	₹14,63,562
Total Cost	₹10,91,160
Total Profit	₹3,72,402

Overall Profit Margin Approximately 25%

The business is profitable overall, but profitability differs considerably between products and discount levels.

12. SALES FORECASTING

Forecast Objective

The purpose of forecasting was to provide a short-term view of expected future sales.

The original sales dates were not evenly spaced because transactions did not occur every calendar day.

This initially prevented Power BI from creating a forecast.

Solution

A continuous Date table was created and connected to the sales data.

The forecast visual was then configured using:

X-axis: Date_Table[Date] → Days

Y-axis: Sales_Data[Sales_Amount]

Analytics → Forecast

Forecast Length: 15 days

This successfully produced the required daily forecast.

Forecast Dashboard



It contains:

- Daily Sales Trend + 15-Day Forecast
- Forecast Matrix
- Monthly Sales Trend
- Sales by Product Category

The forecast should be treated as a short-term planning indicator rather than a highly reliable long-term prediction because the available transaction history is limited.

13. DISCOUNT IMPACT ANALYSIS

A separate business analysis was created to understand the relationship between discounting and business performance.

A Power Query column called **Discount Level** was created.

Classification:

- 0% → No Discount
- Up to 5% → Low Discount
- Up to 8% → Medium Discount
- Above 8% → High Discount

Business Question

Does increasing the discount increase sales enough to justify the reduction in margin?

Visual 1 — SALES BY DISCOUNT LEVEL

This visual compares sales across discount levels.

Results showed:

Discount Level	Sales
Low Discount	₹5,21,265
High Discount	₹4,30,715
Medium Discount	₹3,27,582
No Discount	₹1,84,000

Low Discount generated the highest sales in the dataset.

Visual 2 — PROFIT BY DISCOUNT LEVEL

Profit results:

Discount Level	Profit
Low Discount	₹1,38,705
High Discount	₹95,015
Medium Discount	₹81,982

No Discount ₹56,700

Low Discount generated the highest total profit because it also had considerably more orders.

Visual 3 — MARGIN BY DISCOUNT

Margin results:

Discount Level	Margin
----------------	--------

No Discount	30.82%
-------------	--------

Low Discount	26.61%
--------------	--------

Medium Discount	25.03%
-----------------	--------

High Discount	22.06%
---------------	--------

The analysis shows a clear trade-off:

Higher Discount → Lower Margin

However, lower margin does not automatically mean lower total profit.

14. SQL BUSINESS ANALYSIS

SQL was used for structured business-level aggregation rather than simply repeating Power BI calculations.

Discount Performance

The SQL analysis showed:

Discount Level	Orders	Sales	Profit	Quantity
Low Discount	28	₹5,21,265	₹1,38,705	108
High Discount	15	₹4,30,715	₹95,015	50
Medium Discount	11	₹3,27,582	₹81,982	40
No Discount	10	₹1,84,000	₹56,700	29

Finding

Low Discount had the highest:

- Sales
- Profit
- Orders
- Quantity sold

But this does **not** prove that low discounts caused higher sales.

Product mix, customer mix, and other business factors may influence the result.

15. AVERAGE ORDER PERFORMANCE

SQL was also used to compare performance per order.

Discount Level	Avg Sales / Order	Avg Profit / Order
----------------	-------------------	--------------------

Medium Discount	₹29,780.18	₹7,452.91
-----------------	------------	-----------

High Discount	₹28,714.33	₹6,334.33
---------------	------------	-----------

No Discount	₹18,400.00	₹5,670.00
-------------	------------	-----------

Low Discount	₹18,616.61	₹4,953.75
--------------	------------	-----------

Finding

Medium and High Discount orders generated higher average sales and profit per order.

Low Discount, however, generated the highest overall business contribution because it had substantially more orders.

This demonstrates why both **per-order performance and total business volume** should be considered.

16. ORDER QUANTITY ANALYSIS

Average quantity per order:

Discount Level	Avg Quantity / Order
----------------	----------------------

Low Discount	3.86
--------------	------

Medium Discount	3.64
-----------------	------

High Discount	3.33
---------------	------

No Discount	2.90
-------------	------

Discounted orders were associated with larger baskets than no-discount orders.

However, this is an observed relationship and does not establish that the discount itself caused customers to buy more.

17. PYTHON / PANDAS ANALYSIS

Python was used for a deeper product-level analysis.

Business Question

Which products generate high sales but relatively low profit margins?

Why This Analysis Was Important

Power BI showed overall product performance, while Python was used to examine the relationship between:

Sales + Profit + Margin

at product level.

This provided a more detailed view of product profitability efficiency.

18. PRODUCT-LEVEL PYTHON RESULTS

Product	Sales	Profit	Margin
---------	-------	--------	--------

Laptop	₹3,36,440	₹70,440	20.94%
Monitor	₹2,23,080	₹47,580	21.33%
Tablet	₹1,81,440	₹48,440	26.70%
Desk	₹1,39,750	₹41,350	29.59%
Office Chair	₹1,36,800	₹39,800	29.09%
Printer	₹1,07,880	₹27,080	25.10%
Headphones	₹86,310	₹26,510	30.71%
Keyboard	₹63,140	₹15,140	23.98%
Webcam	₹61,950	₹17,150	27.68%
Mouse	₹53,460	₹12,900	24.13%
Backpack	₹40,832	₹14,532	35.59%
Usb Hub	₹32,480	₹11,480	35.34%

Key Finding

Laptop generated the highest sales but had the lowest product margin at approximately **20.94%**.

Monitor also generated very high sales while having a relatively low margin of approximately **21.33%**.

In contrast, Backpack and USB Hub had margins above **35%**, but their sales volumes were much lower.

This shows that:

High sales ≠ high margin

and

High margin ≠ high total profit

Both dimensions need to be considered together.

19. KEY BUSINESS INSIGHTS

Insight 1 — The Business Is Profitable

The business generated approximately:

₹14.64 lakh Sales

and

₹3.72 lakh Profit

with an overall margin of approximately **25%**.

Insight 2 — Laptop Is the Major Revenue Driver

Laptop generated the highest sales at approximately **₹3.36 lakh**.

However, its margin was only **20.94%**, making it an important product for margin improvement.

Insight 3 — Monitor Also Requires Profitability Attention

Monitor generated approximately **₹2.23 lakh** in sales but had a margin of approximately **21.33%**.

Because of its high sales volume, even a small improvement in margin could have a meaningful effect on total profit.

Insight 4 — High-Margin Products Have Lower Sales

Backpack and USB Hub produced margins above **35%**, but their sales contribution was relatively small.

This suggests an opportunity to explore whether these products can be promoted more effectively.

Insight 5 — Discounts Create a Profitability Trade-Off

As discount levels increase, margin decreases:

30.82% → 26.61% → 25.03% → 22.06%

However, discounted transactions were associated with higher sales volume and larger baskets.

Therefore, discount effectiveness should be judged using **total profit and incremental demand**, not margin percentage alone.

Insight 6 — Sales Volume Matters

Low Discount generated the highest total profit because it generated the highest number of orders.

This demonstrates that a lower-margin strategy can still generate strong total profit when sales volume is sufficiently high.

Insight 7 — Discount Does Not Automatically Mean Causation

The dataset shows an association between discounts and sales/order quantity.

It does not prove that discounts alone caused customers to purchase more.

Product type, customer type, pricing, and other factors may also influence demand.

Insight 8 — Forecasting Supports Short-Term Planning

The 15-day sales forecast provides a forward-looking indicator that can support:

- Inventory planning
- Sales planning
- Promotional decisions
- Staffing
- Short-term business planning

20. BUSINESS RECOMMENDATIONS

Recommendation 1 — Review Pricing of High-Sales Products

Laptop and Monitor should receive particular attention because they generate substantial sales but operate at relatively low margins.

Management should investigate:

- Supplier cost
- Purchase price
- Selling price
- Discount levels
- Product-level costs

Recommendation 2 — Use Discounts Selectively

Discounts should not be applied equally across all products.

The business should identify products where discounts genuinely increase demand and distinguish them from products that customers are likely to purchase without a discount.

Recommendation 3 — Evaluate Total Profit Alongside Margin

Management should avoid making decisions based only on margin percentage.

A lower-margin product can still generate greater total profit when it produces substantially more sales.

The correct evaluation should consider:

Sales + Profit + Margin + Order Volume + Quantity

Recommendation 4 — Promote High-Margin Products

Products such as Backpack and USB Hub have strong margins but lower sales.

Possible strategies include:

- Bundling with high-selling products
- Cross-selling
- Product recommendations
- Targeted promotions

The objective should be to increase their sales without unnecessarily reducing their strong margins.

Recommendation 5 — Monitor High-Discount Products Carefully

High discounts should be monitored because the margin falls to approximately **22.06%**.

Promotions should be continued where the additional business volume justifies the margin sacrifice.

Recommendation 6 — Improve Margin on High-Volume Products

Because Laptop and Monitor have high sales volume, even a modest margin improvement could generate meaningful additional profit.

This may be more valuable than simply increasing sales through deeper discounts.

Recommendation 7 — Use Forecasting for Short-Term Planning

The 15-day forecast should be compared regularly with actual sales.

As additional historical data becomes available, forecasting can become more useful and reliable.

21. FINAL CONCLUSION

This project demonstrates a complete end-to-end data analytics workflow.

The analysis started with messy raw e-commerce data and progressed through:

Data Cleaning → Data Transformation → Data Modeling → Dashboard Development → Forecasting → SQL Analysis → Python Analysis → Business Insights → Recommendations

The most important business lesson from the project is that **sales, profit, and margin must be evaluated together**.

High-sales products can have relatively weak margins, while high-margin products may contribute less total profit because of lower sales volume.

Similarly, higher discounts can reduce margin while potentially increasing transaction volume.

Therefore, the business should focus on **profitable growth rather than sales growth alone**.

The analysis recommends protecting high-volume revenue drivers, improving margins on major products, using discounts selectively, developing high-margin products, and using forecasting to support short-term planning.

22. PROJECT DELIVERABLES

The completed project includes:

Excel

- Raw data inspection
- Initial data validation

Power Query

- Data cleaning
- Date transformation
- Numeric conversion
- Text standardization
- Customer merge
- Product merge

- Target transformation
- Discount classification

Power BI

- Data model
- KPI dashboard
- Sales analysis
- Profitability analysis
- Product analysis
- Regional analysis
- Channel analysis
- Forecast dashboard
- 15-day daily sales forecast

DAX

- Total Sales
- Total Profit
- Profit Margin
- Total Orders
- Total Quantity Sold
- Average Order Value
- Profit

SQL

- Overall sales/profit analysis
- Discount-level analysis
- Average order analysis

- Quantity-per-order analysis

Python/Pandas

- Product sales analysis
- Product profit analysis
- Product margin analysis
- Identification of high-sales/low-margin products

Business Analysis

- Discount impact
 - Sales-volume vs margin trade-off
 - Product profitability
 - Forecast interpretation
 - Business recommendations
-

23. PROJECT VALUE

This project demonstrates practical skills in:

- Data cleaning
- ETL
- Data transformation
- Data modeling
- SQL
- Python/Pandas
- Power BI
- DAX
- Forecasting

- Business analysis
- Data-driven decision making

More importantly, the project demonstrates the ability to move from **raw business data to actionable business recommendations**, rather than only creating charts and calculating KPIs.